



Proposal for

FixGo

.

HerTech Squad -PRP-FixGo-06.06

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1 Introduction

In today's fast-paced digital landscape, both individuals and businesses increasingly seek quick, reliable access to skilled service providers. However, finding trustworthy professionals across various domains from home maintenance to corporate services remains a fragmented and time-consuming process. FixGo addresses this gap by offering a unified web-based platform that simplifies how users discover, evaluate, and connect with service providers across Sri Lanka.

FixGo serves as a comprehensive service marketplace, allowing general customers and businesses to seamlessly browse, book, and manage a wide range of services through a modern and user-friendly interface. The platform supports diverse service categories such as Plumbing Works , Electrical Works , Welding Works , TV Mechanic Services , AC Mechanic Services, Refrigerator Repair Services ,Washing Machine Repair ,Carpentry Services, Gas Stove / Hob Repair, CCTV Installation & Repair , RO / Water Purifier Service and more enabling users to search by location, compare by price and ratings, and request real-time quotations.

What makes FixGo different from other platforms is its focus on automation, transparency, and user convenience. It includes useful features like a booking calendar, an emergency request button, and company project posting options for business clients. To enhance communication, we are also integrating an AI-powered chatbot that can guide users, help with booking, and answer common questions instantly. Service providers can build their own profiles with portfolio images and service history, and customers can leave ratings and reviews after each job.

The system also has an admin dashboard that helps in managing complaints, verifying providers, and generating reports for better monitoring. Secure payment gateway integration and email notifications keep both parties informed throughout the service process. Overall, FixGo aims to provide a smoother and more trusted way for Sri Lankans to hire professionals bringing all service needs under one digital roof.

1.1 Problem statement

Despite the growing demand for digital services, many individuals and businesses in Sri Lanka still face difficulties in finding reliable, verified, and easily accessible service providers across a wide range of categories. Existing solutions are either too general (like online classifieds) or limited in scope, often lacking features such as real-time availability, structured comparison, or proper support mechanisms. Users are forced to rely on manual browsing, unverified contacts, or word-of-mouth recommendations, which can result in delays, poor service quality, and a lack of trust. Moreover, there is no single platform that caters to both general customers and businesses while offering modern functionalities such as price filtering, calendar-based bookings, digital quotations, or emergency service options.

The absence of an intelligent support system further complicates the user journey, especially for non-technical users seeking quick assistance. Service providers, on the other hand, struggle to promote their services effectively, manage bookings, or engage with potential clients in a professional and organized manner. Businesses looking to hire professionals for short-term tasks or specialized projects also lack a structured digital solution to post requirements and manage applications. These gaps highlight the need for a centralized, feature-rich, and intelligent service platform. FixGo aims to solve this problem by offering a streamlined web application that includes verified listings, AI chatbot support, location-based search, quotation generation, scheduling, rating and review systems, and secure payments all within a single user-friendly interface.

1.2 Aim

The aim of this project is to develop a comprehensive and intelligent web-based platform, FixGo, that enables general customers and businesses to efficiently discover, evaluate, and connect with verified service providers across Sri Lanka. The system will simplify the entire service lifecycle from discovery to booking and

payment while offering intelligent support through features such as an AI-powered chatbot, real-time availability tracking, and secure online transactions.

1.4 Objectives

The specific objectives of the FixGo project are:

1. To develop a centralized web application that connects customers and businesses with service providers across multiple categories.
2. To implement location-based search and filtering by service type, pricing, ratings, and availability.
3. To allow service providers to maintain structured profiles, including portfolio samples, ratings, and service details.
4. To provide real-time quotation generation and booking, based on the availability of service providers.
5. To integrate an AI-powered chatbot that assists users with navigation, booking, FAQs, and platform guidance.
6. To enable emergency service requests, allowing users to quickly access urgent services.
7. To support company-level features, including project posting, application management, and provider hiring workflows.
8. To ensure secure payment handling using a payment gateway for smooth and safe transactions.
9. To build an admin dashboard for user verification, complaint handling, and system monitoring.

10. To use email-based push notifications to inform users about bookings, quotations, approvals, and updates.

1.5 Purpose of the Project

The purpose of FixGo is to create a complete digital system that solves the problems mentioned above. Our aim is to:

- Make it easier for customers to find the right provider quickly.
- Allow users to book, chat, and pay all in one place.
- Help service providers grow their business and work more professionally.
- Give companies a platform to post jobs and hire short-term service workers.
- Make the overall service process more transparent and trusted.
- Offer Real-Time Updates and Notifications
- Support fair pricing by allowing users to compare service charges and request quotations.
- Offer emergency help features to handle urgent service needs quickly.
- Ensure service quality by verifying provider details and keeping track of complaints.

2 Literature Survey

2.1 Research Gap

- Most existing research on digital home service platforms focuses on basic features like service listings, manual booking, and provider ratings. However, these systems often lack intelligent support, real-time interaction, and emergency handling.
- They also fail to provide a centralized solution that combines AI assistance, instant quotations, verified provider portfolios, and administrative complaint resolution—especially tailored for developing countries like Sri Lanka.
- This creates a gap that the **FixGo system** aims to address by offering a more intelligent, responsive, and trustworthy platform for home service management.

2.2 Relevant Research Paper

No.	Title	Focus Area
1	An Android Application to Manage House Rental and Maintenance in Sri Lanka	Connecting tenants with handymen via mobile app; focused on streamlining rental & maintenance requests in the local context.
2	Digital Platforms and Women’s Work in Sri Lanka and India	Examining how women engage with home-service digital platforms in Sri Lanka; discusses fee structures, performance, and inclusion.
3	After-sales Services of Home Appliances: Evidence from Sri Lanka	Focuses on customer satisfaction and expectations regarding repair and maintenance services after appliance purchases.

4	Smart GSM-Based Home Automation System	Explores remote control and monitoring of home appliances using SMS in developing countries like Sri Lanka; highlights connectivity challenges.
5	Personal and Household Service Quality Project	Investigates working conditions, service quality, and challenges in the personal and household service sector focus including cleaning and care work

2.3 Research Question

Main Research Question:

How can a web-based intelligent platform enhance the efficiency, trust, and accessibility of household maintenance and service management for residential users in Sri Lanka?

Sub-Questions:

1. What are the key challenges faced by Sri Lankan households in accessing reliable service providers through traditional and existing digital methods?
2. How can AI-powered chatbots improve user experience and reduce service request friction in household maintenance platforms?
3. What features (e.g., real-time quotation, verified portfolios, emergency booking) most influence trust and adoption among urban and semi-urban users?
4. To what extent do platform-based service quality and provider transparency affect customer satisfaction and repeat usage in Sri Lanka?
5. How can web-based complaint handling and administrative dashboards improve post-service support and accountability for service platforms?

2.4 Research Objective

- To implement a user-friendly web-based platform (FixGo) that connects Sri Lankan households with verified service providers for home maintenance tasks.
- To integrate an AI-powered chatbot that assists users in service booking, emergency requests, and general inquiries in real-time.
- To enable features such as real-time quotations, provider profiles with ratings, and booking calendars to increase transparency and trust.
- To improve user satisfaction and reliability by implementing a complaint management system and admin dashboard for service monitoring.
- To evaluate the platform's effectiveness in enhancing service accessibility and reducing delays through usability testing and user feedback.

3 Proposed Approach

3.1 Development Approach

The development of the FixGo web application follows a modular, layered architecture guided by the Agile methodology, ensuring iterative development, continuous feedback, and adaptability to changes. The project is divided into multiple sprints, each focused on building and integrating core modules such as user authentication, service listing, booking engine, AI chatbot, and admin dashboard.

Software Development Methodology

The team adopts the Agile Scrum approach, which includes:

- Sprint planning meetings
- Updates through the Slack and whatsapp
- Iterative releases with sprint reviews
- Continuous testing and integration

This methodology ensures quick adaptability to evolving requirements and frequent validation from end-users and stakeholders.

Technology Stack

- Frontend: ReactJS
- Backend: Spring Boot
- Database: PostgreSQL
- Authentication: JWT
- AI Chatbot
- Notifications: Email alerts via SMTP and integrated modules
- Deployment: GitHub

Layered Architecture

1. Presentation Layer: ReactJS interface for users, service providers, and admin
2. Logic Layer: Core services such as booking, quotation, emergency requests, AI chatbot, and payment handling
3. Data Layer: PostgreSQL for storing users, services, quotations, reviews, and availability

Tools & Platforms

- Version Control: Git & GitHub
- Task Management: Trello
- Collaboration: Google Meet, Figma, Draw.io , Slack
- Testing: Postman, JUnit, and Jest

Security Considerations

- Input validation and sanitization
- Password hashing
- Role-based access control (RBAC)
- Secure API endpoints with HTTPS and token-based authentication

3.2 Project Scope

- Registration and login for customers, service providers, companies, and admin
- Service listing, searching, filtering, and booking
- Quotation requests and responses
- Emergency request button
- AI chatbot support
- Email notifications for updates
- Payment gateway for secure transactions
- Forum for discussion and help
- GPS tracking for service location
- Company job posting and application handling
- Admin panel for monitoring and complaint handling

4 Project Plan

Assume that the project will begin on May 20th. You can include a Gannt chart even.

Stage	Duration	Start	Target End
Requirement Gathering & Analysis	2 weeks	20-May-2025	02-Jun-2025
Competitor Study & Feature Mapping	1 week	03-Jun-2025	09-Jun-2025
UI/UX Design (Wireframe + Prototype)	2 weeks	10-Jun-2025	23-Jun-2025
Database Design & Schema Setup	1 week	24-Jun-2025	30-Jun-2025
Backend API Development (Spring Boot)	3 weeks	01-Jul-2025	21-Jul-2025
Frontend Development (ReactJS)	3 weeks	22-Jul-2025	11-Aug-2025
AI Chatbot Integration (Python API)	2 weeks	12-Aug-2025	25-Aug-2025
Booking, Quotation & Emergency Flow	2 weeks	26-Aug-2025	08-Sep-2025
Payment Gateway & Email Notification	1.5 weeks	09-Sep-2025	19-Sep-2025
Admin Panel + Provider Management	2 weeks	20-Sep-2025	03-Oct-2025
Unit Testing + Debugging Phase	2 weeks	04-Oct-2025	17-Oct-2025
Integration Testing & UAT	2 weeks	18-Oct-2025	31-Oct-2025
UI Improvements & Final Touches	2 weeks	01-Nov-2025	14-Nov-2025
Deployment to Hosting Platform	1 week	15-Nov-2025	21-Nov-2025
Documentation & Final Report	4 weeks	22-Nov-2025	19-Dec-2025
Final Review, Corrections & Viva Preparation	6 weeks	20-Dec-2025	31-Jan-2026

4.1 Assumptions and Constraints

Assumptions

1. User Internet Access: It is assumed that all end users (customers, service providers, admins) have access to a stable internet connection for interacting with the web platform.

2. **Browser Compatibility:** The system will be accessed primarily through modern web browsers such as Google Chrome, Firefox, or Microsoft Edge.
3. **Third-party APIs Availability:** External services (e.g., email notification APIs, payment gateways, and AI chatbot platforms) will remain available and functional during development and operation.
4. **Defined Roles and Permissions:** The roles of users (e.g., customer, service provider, admin, advisor) are clearly defined and will not change drastically during development.
5. **Team Availability:** The development team will be available as per the project timeline and milestones for continuous development, testing, and deployment.
6. **Feature Scope Agreement:** The project scope and features discussed are agreed upon and any additions will be handled via change requests.

Constraints

1. **Technology Limitation:** The project is limited to the chosen stack — ReactJS for frontend, Spring Boot for backend, PostgreSQL as the database, and AI chatbot service.
2. **Time Constraint:** The project must be completed and submitted by April 1st, 2026, including final report, presentation, and evaluation.
3. **Resource Constraints:** The team has limited access to premium services (e.g., paid hosting, advanced APIs) and will rely mostly on open-source or free-tier tools.
4. **AI Chatbot Scope:** The chatbot will handle basic user queries and booking assistance but will not include advanced natural language learning or voice interaction.
5. **No Mobile App:** The project will be a responsive web application only; no native mobile app will be developed.

6. Data Privacy and Security: The platform must ensure basic security practices (authentication, input validation, role access), but advanced security like encryption at scale or GDPR compliance is outside this version’s scope.

4.2 **Work Distribution** .

Team Member	Key Responsibilities
K. Kethusha	Backend development (Spring Boot APIs), Admin dashboard, Payment integration, Testing & QA
P. Mathushi	Frontend UI development (ReactJS), Booking & Quotation module, Email notifications
M. Kosika	AI Chatbot development, API integration, Service category logic, Competitor research
A. Ajanthika	Database design (PostgreSQL), Emergency request system, Provider module development
E. Sanjeeveni Aishwarya Devi	Frontend enhancements, Admin interface support, AI Chatbot development, Documentation, Competitor research

4.3 **Project Management Strategies**

As a team, we adopt a lightweight Agile approach that supports flexibility, transparency, and continuous improvement throughout the project lifecycle.

- Weekly Team Meetings:
We conduct weekly meetings (online or in person) to plan, review progress, and resolve any blockers collaboratively.

- **Task Management Using Trello:**
All tasks are assigned and tracked using Trello boards, ensuring accountability and visibility for each team member.
- **Shared Google Calendar:**
Key dates such as task deadlines, review meetings, and submission milestones are managed in a shared calendar accessible to the whole team.
- **Version Control with GitHub:**
Our codebase is hosted on GitHub to support team collaboration, code reviews, backup, and version tracking.
- **Milestone-Based Planning:**
The project is divided into major phases—Proposal, Prototype, Development, Testing, and Final Submission—each with clear objectives and timelines.

4.4 Risk Management Plan

Although this is an academic project, we are aware of possible challenges and have outlined strategies to minimize their impact:

Potential Risk	Impact	Likelihood	Mitigation Strategy
Conflicts with academic deadlines	Reduced project attention	Medium	Align project tasks with academic calendar and plan ahead.
Team member falling behind on tasks	Delays in progress	Medium	Regular meetings and team support to redistribute workload if needed.
Technical difficulties or skill gaps	Development challenges	High	Use online tutorials, peer help, and guidance from lecturers.
Poor communication	Misunderstandings or missed work	Low	Maintain communication through WhatsApp/Discord and weekly updates.

Last-minute changes in requirements	Extra pressure near deadlines	Medium	Document requirements early and get feedback from the supervisor.
Technical Skill Gaps	Slower development	Medium	Allocate time for upskilling; peer learning sessions.

5 Cost Proposal

The project follows a Time and Materials pricing model, simulating real-world development costs using estimated student-based labor and basic infrastructure expenses. This includes development efforts, documentation, and essential operational costs such as hosting and third-party service APIs.

Estimated Development and Infrastructure Cost

Item / Activity	Total Cost (LKR)
UI/UX Design	10,000
Frontend Development (ReactJS)	25,000
Backend Development (Spring Boot)	36,000
AI Chatbot Integration	15,000
Database Design & Setup	10,000
Testing & QA	10,000
Project Management & Documentation	12,000
Domain Name (e.g., .lk or .com)	3,500
Hosting (Render, Firebase, etc.)	6,000
AI Chatbot API Usage	5,000

Total Estimated Project Cost: LKR 152,500

6. Conclusion

FixGo is a comprehensive digital platform developed to address the challenges faced by customers and service providers in today's service-driven economy. The platform offers an organized and user-friendly solution for discovering, booking, and managing services in one place. By incorporating key features such as service listings, secure online payments, real-time status updates, and role-based user access, FixGo enhances the overall experience for both individual customers and service professionals.

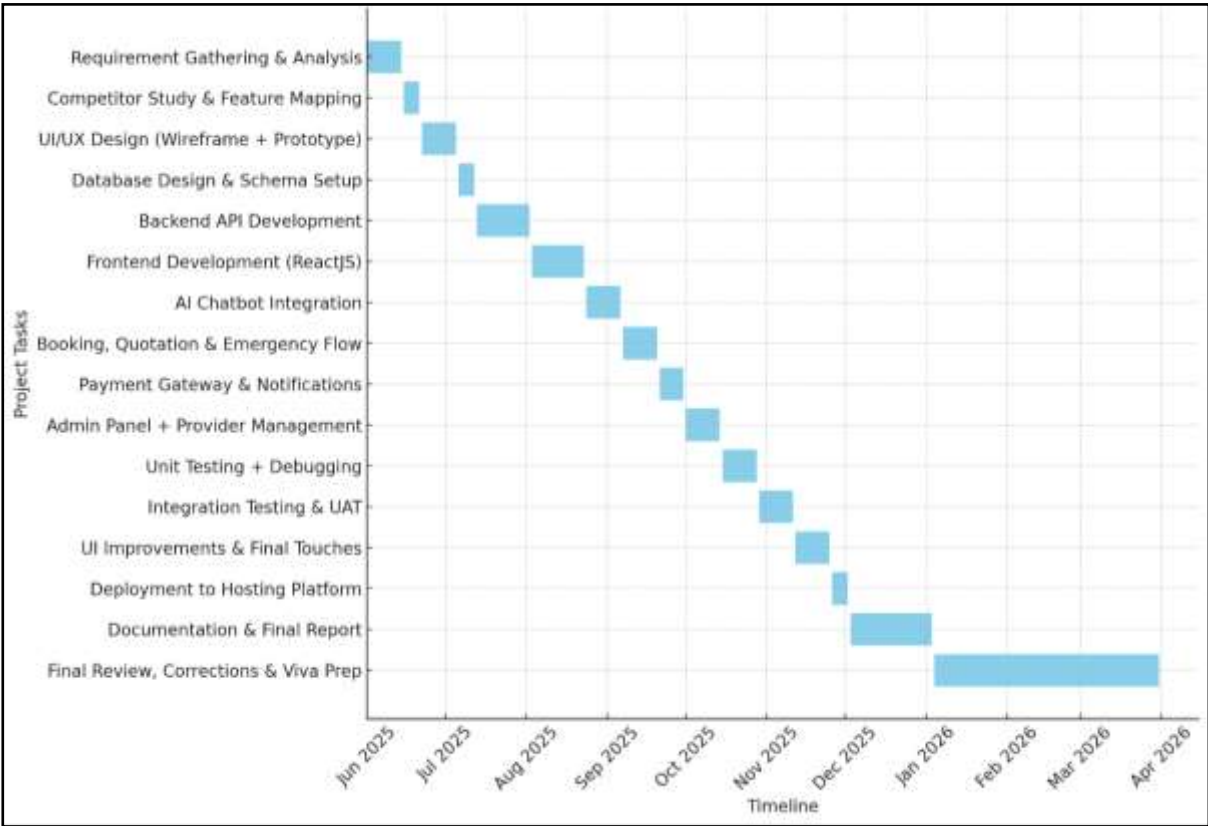
This system aims to reduce the time, effort, and confusion typically associated with finding and managing trusted services. Whether it's for household maintenance, freelance work, or short-term contract jobs, FixGo enables users to interact seamlessly in a secure and transparent environment. The platform supports professionalism, improves customer satisfaction, and helps service providers expand their visibility and credibility in the market.

As development progresses, the FixGo team is committed to building a robust and scalable system that not only delivers on its core objectives but also lays the foundation for future innovations. Our goal is to create a service ecosystem that is

efficient, reliable, and accessible to all users, ultimately contributing to the digital transformation of everyday service management.

7. Appendix

Gantt Chat of FixGo (June 2025 to Apirl 2026)



Meeting Minutes

<https://docs.google.com/document/d/1VoirlBQCkzVSWX4e1mDzGQZY5gy8wIRb3-8bqVaU4I/edit?usp=sharing>

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Tab 1

Meeting-Heading

Agenda:

Notes/Clarifications:

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Meeting-Heading

Meeting information		
Meeting Date/Time	20 th May 2025 1:30 PM to 3:30 PM	
Participants	K. Kishu	820004051
	P. Anshu	802003309
	M. Kishu	802000743
	A. Anshu	802079057
	E. S. Anshu	802008839
Estimated Time	3 hours	Actual Time
Special Notes	Project Idea, Real world Problem, Scope	
Call/Location Information	Google Meet	
Supported Document	Project Proposal Template	

Agenda:

Topic 1: Overview of FixGo project and its main goals.

Topic 2: Team role assignment and task confirmation.

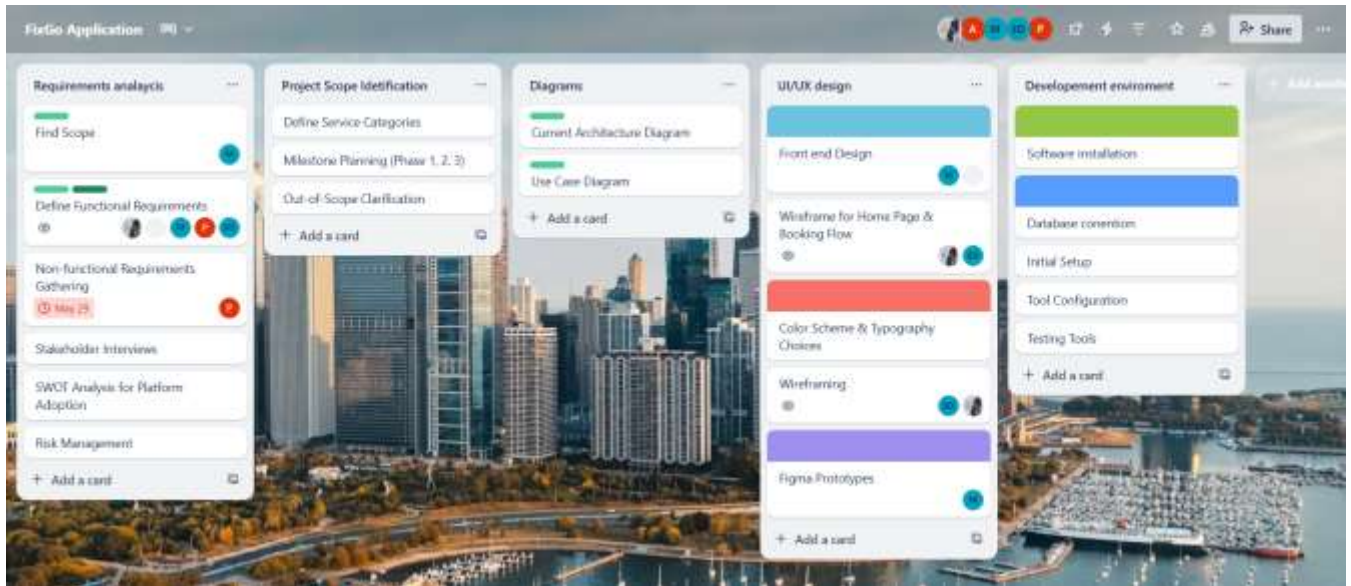
Topic 3: Finalize tools and technologies for development.

Topic 4: Confirm project methodology and timeline structure.

Notes/Clarifications:

All team members acknowledged and approved their responsibilities.

Project Management Tool – Trello Board



8. Reference

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